

# AUSTIN FERRO, PhD

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## EDUCATION

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|----------------|--------------------------------------------------------------------------------|----------------------|
| Fall 2015-2020 | University of Minnesota- Twin Cities<br>Graduate Ph.D. program in Neuroscience | Minneapolis, MN      |
| 2011-2015      | Skidmore College<br>Bachelor of Arts in Neuroscience                           | Saratoga Springs, NY |

## POSITIONS AND APPOINTMENTS

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|----------------------|------------------------------------------------------------------------|------------------------|
| Summer 2025- present | Howard Hughes Medical Institute<br>Research Specialist II: Cheadle Lab | Cold Spring Harbor, NY |
| Fall 2020- 2025      | Cold Spring Harbor Laboratory<br>Postdoctoral Fellow: Cheadle Lab      | Cold Spring Harbor, NY |

## RESEARCH EXPERIENCE

**Postdoctoral Fellow (2020-Present)** Dr. Lucas Cheadle's laboratory, Cold Spring Harbor Laboratory (CSHL), Cold Spring Harbor, NY.

Determining refinement mechanisms of neural circuits during visual cortical circuit development (Auguste\*, **Ferro\***, et al., 2022) and defining the roles of neuro-immune signaling in adult hippocampal neural circuit function (**Ferro** et al., 2026; **Ferro\*** et al., *in prep*). In these works, I independently designed and implemented experiments that discovered oligodendrocyte precursor cells (OPCs) refine visual circuits – a novel function partially mediated by microglia -- during the sensory experience-dependent phase of visual cortical development (Auguste\*, **Ferro\***, et al., 2022). Further, I discovered that adult microglia cytokine signaling via TWEAK-Fn14 acts as an inhibitory brake on neural circuit activity, maintains circadian activity in CA1 pyramidal neurons, and limits the ability of neural circuits to reach pathological activity states (**Ferro**, Vita et al., 2026). In work, currently in preparation, I am determining the extent of microglial regulation of the noncanonical neuron-OPC synapse. To probe microglia-OPC-synapse interactions I have designed multiple tools to quantify cell-cell interactions including a custom 3-channel two-photon microscope, custom acquisition software, viral tools for synaptic labeling, as well as created open source python packages that implement machine learning for analysis. Taken together, my postdoctoral work has uncovered novel mechanisms of neural circuit development and function and has revealed OPCs and microglia as a potential druggable target for various neuropathologies found in neurodevelopmental disorders where refinement is disrupted, such as autism spectrum disorder.

## GRADUATE RESEARCH EXPERIENCE

**PhD Candidate (2015-2020)**. Laboratories of Drs. Marija Cvetanovic and Alfonso Araque. University of Minnesota, Minneapolis, MN. Identification of disease modifying functions of glia in spinocerebellar ataxia type 1 (SCA1).

During my PhD work, I investigated and identified potential roles of both microglia (Ferro et al., 2018, Ferro et al., 2019a, Ferro et al., 2019b) and astroglia (under review) in the pathogenesis of SCA1. To characterize the

function of glia in SCA1, I combined molecular, morphological, and electrophysiological characterization alongside genetic modifications of mouse models in SCA1.

## GRANTS AND FUNDING

2018 PI F31 NS103392-01A1, NIH  
 2018 PI Sping and Ying Ngho Lin Award for Research Independence, University of Minnesota  
 2018 PI NAF Young Investigator grant, National Ataxia Foundation

## HONORS AND AWARDS

2026 Scientific Breakthrough Award, CSHL  
 2025 Lecture selection for SPiNES NYU Neuroscience Institute at New York University  
 2018 1<sup>st</sup> place poster winner Wallin Neuroscience Discovery Day, University of Minnesota  
 2018 Hot Chair travel grant, National Ataxia Foundation  
 2017 Stark Award, University of Minnesota.  
 2015 Faculty for Undergraduate Research Travel Award, Skidmore College.  
 2015 Hartnett Prize for Undergraduate Research, Skidmore College.  
 2013 Top finalist in the Student Poster Competition at the Neurotrauma Symposium, Roskamp Institute.  
 2013 Summer Collaborative Research Grant, Skidmore College.

## PUBLISHED MANUSCRIPTS

\* Authors contributed equally

1. **A. Ferro** \*, D.J. Vita\*, T. Fallon, A. Arshad, L. Boyd, T. Stanley, Q. Lin, A. Berisha, U. Vrudhula, AM. Gomez, I. Sanchez-Martin, JC. Borniger, L. Cheadle. Fn14 is an activity-dependent, Bmal1-regulated cytokine receptor that induces rod-like microglia and restricts neuronal activity in vivo. *Cell Reports*, 45(2):116926. (2026)
2. D. Ford-Roshon, M. Dudek, A. Glynn, A. Glasman, J. York, E. Lawrence, D. Nguyen, L. Shinn, G. Berry, L. Kendall, J. Bonner, **A. Ferro**, S. Lagalwar, Spinocerebellar Ataxia Type 1 (SCA1) Cell Models Display Widespread Mitochondrial and Extra-Nuclear Alterations. *Journal of Molecular Neuroscience.*, 75:4, 1–17 (2025).
3. J. A. Kahng, A. M. Xavier, **A. Ferro**, S. X. Tang, Y. S. S. Auguste, L. Cheadle, High-confidence and high-throughput quantification of synapse engulfment by oligodendrocyte precursor cells. *Nature Protocols* 1–33 (2024).
4. **A. Ferro**\*, Y. S. S. Auguste\*, J. A. Kahng, A. M. Xavier, J. R. Dixon, U. Vrudhula, A. S. Nichitiu, D. Rosado, T. L. Wee, U. V. Pedmale, L. Cheadle, Oligodendrocyte precursor cells engulf synapses during circuit remodeling in mice. *Nature Neuroscience* 25 (2022).
5. A. Mellesmoen, C. Sheeler, **A. Ferro**, O. Rainwater, M. Cvetanovic, Brain Derived Neurotrophic Factor (BDNF) Delays Onset of Pathogenesis in Transgenic Mouse Model of Spinocerebellar Ataxia Type 1 (SCA1). *Frontiers in Cellular Neuroscience* 12, 509 (2019).
6. **A. Ferro**, W. Qu, A. Lukowicz, D. Svedberg, A. Johnson, M. Cvetanovic, Inhibition of NF- $\kappa$ B signaling in IKK $\beta$ F/F;LysM Cre mice causes motor deficits but does not alter pathogenesis of Spinocerebellar ataxia type 1. *PLoS One* 13 (2018).
7. **A. Ferro**, E. Carbone, J. Zhang, E. Marzouk, M. Villegas, A. Siegel, D. Nguyen, T. Possidente, J. Hartman, K. Polley, M. A. Ingram, G. Berry, T. H. Reynolds, B. Possidente, K. Frederick, S. Ives, S. Lagalwar, Short-term succinic acid treatment mitigates cerebellar mitochondrial OXPHOS dysfunction, neurodegeneration and ataxia in a Purkinje-specific spinocerebellar ataxia type 1 (SCA1) mouse model. *PLoS One* 12, e0188425 (2017).
8. **A. Ferro**, E. Carbone, E. Marzouk, A. Siegel, D. Nguyen, K. Polley, J. Hartman, K. Frederick, S. Ives, S. Lagalwar, S. Treating SCA1 mice with water-soluble compounds to non-specifically boost mitochondrial function. *Journal of Visualized Experiments* 119 (2017).

9. B. C. Mouzon, C. Bachmeier, **A. Ferro**, J.-O. Ojo, G. Crynen, C. M. Acker, P. Davies, M. Mullan, W. Stewart, F. Crawford, Chronic neuropathological and neurobehavioral changes in a repetitive mild traumatic brain injury model. *Annals of Neurology* 75, 241–254 (2014).
10. L. Abdullah, J. E. Evans, A. Bishop, J. M. Reed, G. Crynen, J. Phillips, R. Pelot, M. A. Mullan, **A. Ferro**, C. M. Mullan, M. J. Mullan, G. Ait-Ghezala, F. C. Crawford, Lipidomic Profiling of Phosphocholine Containing Brain Lipids in Mice with Sensorimotor Deficits and Anxiety-Like Features After Exposure to Gulf War Agents. *Neuromolecular Medicine* 14, 349–361 (2012).

## MANUSCRIPTS IN PREPARATION

† Corresponding or Co-corresponding author

1. **A. Ferro**<sup>†</sup>, A. Ademola, J. Kahng, C. Zhang, D.J. Vita, G. Chan, Q. Lin, T. Fallon, J. Monteiro, N. McNamara, V. Miron, J. Trottmann, D. Bergles, L. Cheadle<sup>†</sup>, Glia-Glia Crosstalk: Homeostatic control of OPC form and function by microglia. Target Journal: *Nature*. Target submission: Fall of 2026.

Oligodendrocyte precursor cells (OPCs) are the only nonneuronal cell within the central nervous system to receive direct synaptic innervation, yet the mechanisms governing their synaptic and morphological state remain poorly understood. Here we report are essential regulators of OPC synaptic interactions and arbor complexity, suggesting that microglia are potent regulators of the neuron-OPC synapses and of OPC form and function.

2. T. Fallon, L. Cheadle, **A. Ferro**<sup>†</sup>, SNACC: open-source, model-independent quantification of cell–cell interactions and engulfment in three-dimensional tissue volumes. Target Journal: *Nature Methods*. Target submission: Fall of 2026.

**Segmented Neighbor Analysis of Cellular Components (SNACC)** is an open-source, model-independent Python framework for quantifying cell–cell interactions and engulfment from 3D fluorescent volumes. For proof of principle, we trained Cellpose models against oligodendrocyte precursor cells (OPCs), microglia, synaptic material and lysosomal organelles, adapting Cellpose output for the highly ramified morphologies of glia, to assess whether pharmacological inhibition of Trol3, Axl, and MertK receptors reduces glial-phagocytosis of synapses in mouse cortex.

## PUBLISHED REVIEWS

1. D. J. Vita, **A. Ferro**, L. Cheadle, Experience-dependent control of synaptic remodeling and structural plasticity by glia. *Current Opinion in Neurobiology* 93, 103059 (2025).
2. T. E. Faust, B. A. Devlin, I. Farhy-Tselnicker, **A. Ferro**, M. Postolache, W. Xin, Glial Control of Cortical Neuronal Circuit Maturation and Plasticity. *Journal of Neuroscience* 44 (2024).
3. **A. Ferro**, L. Cheadle, When the levee of sympathetic outflow breaks. *Immunity* 55, 1334–1336 (2022).
4. **A. Ferro**, Y. S. S. Auguste, L. Cheadle, Microglia, Cytokines, and Neural Activity: Unexpected Interactions in Brain Development and Function. *Frontiers in Immunology* 12, 2546 (2021).
5. **A. Ferro**, C. Sheeler, J. G. Rosa, M. Cvetanovic, Role of Microglia in Ataxias. *Journal of Molecular biology* (2019). <https://doi.org/10.1016/j.jmb.2019.01.016>.
6. **A. Ferro**, C. Sheeler, M. Cvetanovic, Microglial Self-Recognition STINGs in A-T Neurodegeneration. *Trends in Neurosciences* (2019). <https://doi.org/10.1016/j.tins.2019.09.005>.

## SELECT INVITED TALKS

1. **A. Ferro**. Microglia coordinate OPC-Synapse Interactions. Invited SPiNES seminar talk at New York University (2026), New York, NY.

2. **A. Ferro**. Microglia coordinate OPC-Synapse Interactions. Invited seminar talk at Massachusetts General Hospital Mucosal Immunology and Biology Research Center (2025), Boston, MA.
3. **A. Ferro**. Microglia coordinate OPC-Synapse Interactions. Invited seminar talk at Hospital del Mar Research Institute (2025), Barcelona, Spain.
4. **A. Ferro**. Dynamic interactions between microglia and oligodendrocyte precursor cells (OPCs) remodel neural circuits in response to sensory experience. Minisymposium at SFN (2024), Chicago, IL.
5. **A. Ferro**. Microglia coordinate OPC-Synapse Interactions. Talk at 2024 CSHL Glia, Cold Spring Harbor, NY.
6. **A. Ferro**, J. Kim, A. Araque, M. Cvetanovic. Biphasic function of Bergmann glia in SCA1. Talk at 2018 National Ataxia Foundation Investigators Meeting, Philadelphia, PA.
7. **A. Ferro**, A. Araque, Marija Cvetanovic. Bimodal function of Bergmann glia in SCA1 disease progression. Talk at the 2017 International Astrocyte School, Bertinoro (Italy).

## TEACHING, SERVICE AND MENTORING

### **Reviewer for:**

-Neuroscience

-*Acta Neuropathologica Communications*

-*Cell Reports*

### **Mentorship:**

**Student Mentor (2020-Present).** Laboratory of Dr. Lucas Cheadle. CSHL, Cold Spring Harbor, NY.

Advised and mentored mentees including high school, undergraduate, graduate students, and postdocs in various lab techniques, experimental design, coding, as well as grantsmanship.

Uma Vrudhula- Cheadle Lab technician

Yohan Auguste- Cheadle Lab technician

Anne-Sarah Nichitiu- Partners for the Future scholar, 2021

Jessica Kahng- CSHL School of Biological Sciences PhD candidate, Entering class of 2020

Samantha Tang- Stony Brook University, Department of Neurobiology and Behavior PhD candidate, Entering class of 2021

Anosha Arshad- Stony Brook University, Department of Neurobiology and Behavior Masters Student, Graduated 2024

Aisha Ademola- Cheadle Lab technician

Patrick Werhle- Stony Brook University, Department of Neurobiology and Behavior PhD student, Entering class of 2023

Dom J. Vita, PhD- Cheadle Lab Postdoctoral fellow, 2024

Melissa Glick- Cheadle Lab CSHL Undergraduate Research Program student, 2026

Trevor Fallon- Cheadle Lab CSHL Postbaccalaureate Research Education Program student, 2026

**Student Mentor (2015-2020).** Laboratories of Drs. Cvetanovic and Araque. University of Minnesota, Minneapolis, MN.

Advised and mentored mentees ranging from undergraduate to graduate students in various lab techniques, experimental design, coding, as well as grantsmanship.

Katie Hammel- University of Minnesota GPN, Entering class of 2018

Juao Rosa- Cvetanovic Lab technician

Ella Borgenhimer- Cvetanovic Lab technician

Carlee Toddes- University of Minnesota GPN, Entering class of 2016

Nicole Zarate- University of Minnesota GPN, Entering class of 2017

Marietta Monteivero- Macalester College, Class of 2018

Jack Martin- Colorado College, Class of 2019

**Student Mentor (2013-2015).** Laboratory of Dr. Sarita Lagalwar. Skidmore College, Saratoga Springs, NY.

Advised and mentored college students in various lab techniques as well as experimental design. As a mentor, I taught techniques such as western blotting, confocal microscopy, cell culture, transmission electron microscopy, pathology, and pharmacology to various people in the lab. Advised a high school student, Annie Getz, in designing a cell culture study to test over the counter herbicides on neuroblastoma cell development. Annie earned the Eleanor Miller Reed Science Research Award at the RPI Greater Capital Region Science Fair and submitted her work to the Science Intel-ISEF science competition.

Chloe Larson- Skidmore College, Class of 2016

Jenny Zhang- Skidmore College, Class of 2016

Monica Villegas- Skidmore College, Class of 2016

Porter Hall- Skidmore College, Class of 2016

Annie Getz- Saratoga Springs High School, Class of 2016

### **Teaching Service:**

**Guest Lecture (2019).** Neuroscience concentration capstone course (NEUR395). Carleton College, Northfield, MN.

**Glia Journal Club President (2019).** I directed a biweekly journal club dedicated to the function of Glia in healthy and pathological states. University of Minnesota, Minneapolis, MN.

**Guest Lecturer (2018).** Cellular Signaling in the Nervous System: The Old and New Players. Macalester College, St. Paul, MN.

**Teaching Assistant (2017).** Introduction to Neurobiology II: Perception and Behavior. University of Minnesota, Minneapolis, MN.

### **ADDITIONAL LINKS**

OrcID: <https://orcid.org/0000-0001-7359-395X>

Google Scholar: <https://scholar.google.com/citations?hl=en&user=997G3cYAAAAJ>

GitHub: <https://austinferro.github.io/>

LinkedIn: <https://www.linkedin.com/in/austin-ferro-phd-b4638987/>

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